Reg. No. | | | | | | | | | | |

[14ME653]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE – 2019
HYDRAULICS AND PNEUMATIC SYSTEMS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
1. Answer any **FIVE** full questions, choosing at least TWO questions from each part.
2. Missing data, if any, may be suitably assumed.

PART-A

1. a. Explain the construction and working of balanced vane pump. **06**

b. Explain the construction and working of internal gear pump. **08**

2. a. Explain the operation of a gear motor with a neat sketch. **06**

b. A hydraulic motor has a displacement of 165 cm^3 and operates with a pressure of 70bar and speed of 2000rpm. If the actual flow rate consumed by the motor is 6LPS (litre per second) and actual torque delivered by the motor is 170N-m, find **08**

i. Volumetric efficiency

ii. Mechanical efficiency

iii. Overall efficiency

iv. Actual power delivered by motor.

3. a. Explain with a neat sketch the working of four-way, two position direction control valve with symbolic representation. **06**

b. Explain with a neat sketch working of a simple pressure relief valve. **08**

4. a. Explain with suitable circuits, how single acting and double acting cylinders are controlled. **08**

b. Sketch and explain the working of a locked cylinder hydraulic circuit using pilot check valves. **06**

PART-B

5. a. Write a short note on hydraulic seals. **08**

b. List and explain desirable properties of hydraulic oils. **06**

6. a. State the advantages of pneumatic systems. **06**

b. What is a rodless cylinder? Explain any two types of rodless cylinders. **08**

7. a. Explain with a neat sketch construction and operation of a typical quick exhaust valve to increase the actuation speed of a cylinder in a pneumatic circuit. **08**

b. With pneumatic circuit, explain the creation of AND logic. **06**

8. a. Explain with a neat diagram coordinated sequence motion of two cylinders. **08**

b. Describe the different methods to design multi-cylinder circuits. **06**

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